



C23-AEI-BM-CH-CHST-CHOT
-CHPC- CHPP-EE-EEVT-TT-102

23006

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

FIRST YEAR (COMMON) EXAMINATION

ENGINEERING MATHEMATICS—I

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **three** marks.

1. If $A = \{-1, 0, 1\}$ and $f : A \rightarrow B$ is a function such that $f(x) = x^2 + 1$, then find the range of f .
2. Resolve $\frac{x}{(x+1)(x-8)}$ into partial fractions.
3. If $A = \begin{bmatrix} & - \\ & \end{bmatrix}$ and $B = \begin{bmatrix} & & \\ - & & \\ & - & \end{bmatrix}$, then find $A - B$.
4. If $A = \begin{bmatrix} & \\ & \end{bmatrix}$, then find $\text{adj } A$.
5. Prove that $\frac{\cos 6^\circ - \sin 6^\circ}{+} =$
6. Find the value of $\sin 15^\circ$

7. Find the modulus of the complex number $(3+i)(2-7i)$.
8. Find the equation of the straight line passing through the points $(1, 2)$ and $(3, -4)$.
9. Evaluate $\lim_{x \rightarrow 0} \frac{\sin 5x}{\sin 3x}$.
10. Find the derivative of $5x^2 - 9x + 2$ w.r.t. x .

PART—B

10×5=50

Instructions : (1) Answer *any five* questions.
(2) Each question carries **ten** marks.

11. Solve the equations by using Cramer's rule : $3x + 2y + 2z = 5$,
 $2x + y - z = 1$ and $x + y - z = 0$.
12. (a) Prove that $\sin 50^\circ - \sin 70^\circ + \sin 10^\circ = 0$
(b) Prove that $\tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{5}\right) = \tan^{-1}\left(\frac{3}{11}\right)$
13. (a) Solve $2\cos^2 \theta - 3\cos \theta + 1 = 0$.
(b) Find the area of the $\triangle ABC$ when $a = 3$, $b = 6$, $c = 7$.
14. (a) Find the equation of the circle having $(1, 2)$ and $(-2, 3)$ as end the points of its diameter.
(b) Find the equation of the parabola whose focus is $(5, 0)$ and directrix is $x + y + 1 = 0$.

15. (a) Find the derivative of $3 \sin x - \cos x + \log x$ with respect to x .
(b) Find the derivative of $x \log x$ with respect to x .
16. (a) If $y = x^x$, then find $\frac{dy}{dx}$.
(b) If $y = a \cos x + b \sin x$, then prove that $\frac{d^2y}{dx^2} + y = 0$.
17. (a) Find the slopes of the tangent and normal to the curve $y = x^2 + 2$ at $(0, 2)$.
(b) If $s(t) = t^2 + 2t + 3$, is a displacement of a particle, then find its velocity and acceleration at $t = 3$ sec.
18. (a) Find the maximum or minimum value of the function $f(x) = (x - 2)(2x + 1)$.
(b) The side of a square plate is measured as 5 cm instead of 5.03 cm. Find approximate percentage error in its perimeter.

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